pMSSM Parameter Space Investigation and ATLAS Blind Spot Analysis

J. Wuerzinger Claude (Anthropic)

pMSSM Working Group Meeting

March 2026

Outline

1. Introduction
2. Scan Strategy & Results
3. ATLAS Coverage
4. ATLAS Blind Spots
5. Complementarity & Recommendations

Motivation & Method

Why a pMSSM scan?

- Simplified models assume 100% BRs — pMSSM does not
- 19-parameter scan → realistic spectra with mixed decays, compressed regions, co-annihilation corridors
- Goal: identify **actual** ATLAS blind spots and propose Run 3 searches

8-step pipeline:

SPheno → Softsusy → micrOMEGAs → SuperISO → GM2Calc → FeynHiggs → SModelS 3.1.1 → Ntuple

Pipeline details

100% AI-driven project

Every component — scan design, code, analysis, plots, and these slides — was produced by **Claude** (Anthropic) via Claude Code.

Full codebase:

`https://github.com/jwuerzinger/SUSY_agent` **6**

selection cuts:

Higgs mass	$122 < m_h < 128$ GeV
Relic density	$\Omega h^2 \leq 0.132$
B-physics	2σ
SModelS	$r < 1$
LEP	$m(\tilde{\chi}_1^{\pm}) > 103$ GeV
Quality	SP + SS converge

Cut details

Scan Strategy & Dataset

Five-phase scan:

1. Flat random (2 k pts $\rightarrow$ 5)
2. Targeted MCMC $\times$4 chains (7 k $\rightarrow$ 451)
3. Refinements (4.5 k $\rightarrow$ 2)
4. Blind-spot targeting (7.1 k $\rightarrow$ 1,605)
5. Importance sampling (4.8 k $\rightarrow$ 24)

26,048 evaluated $\rightarrow$ 2,087 pass all cuts

Phase breakdown

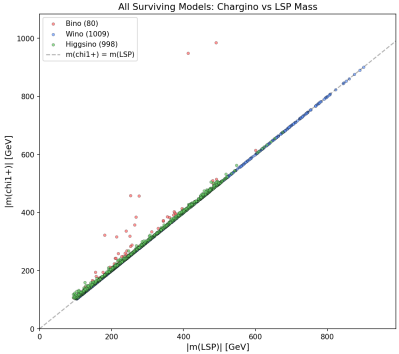
LSP composition:

Type	N	Frac.
Wino	1,009	48.3%
Higgsino	998	47.8%
Bino	80	3.8%

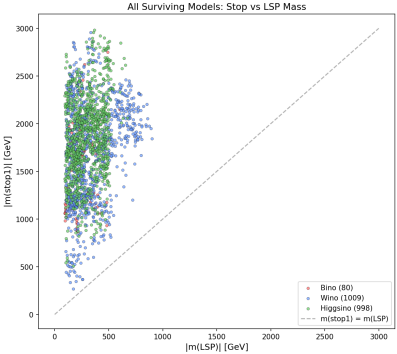
- Wino/Higgsino: naturally under-abundant
- Bino: needs co-annihilation or A-funnel (narrow corridors)
- 99.5% have $\Delta m(\tilde{\chi}_1^{\pm}, \mathrm{LSP}) < 50\,\mathrm{GeV}$

Category plot

Surviving Model Landscape



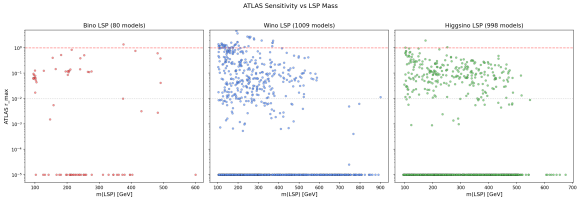
Chargino vs LSP mass. Models cluster along the diagonal — compressed spectra dominate.



Stop vs LSP mass. 293 light stops, only ∼2% excluded. 7 compressed stops at $\Delta m < 200\,\mathrm{GeV}$.

▸ Categories plot

ATLAS Coverage (SModelS v3.1.1)

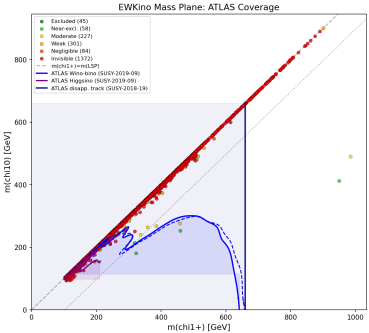


Tier	N	Frac.
Invisible	1,372	65.7%
Negligible	84	4.0%
Weak	301	14.4%
Moderate	227	10.9%
Near-excl.	58	2.8%
Excluded	45	2.2%

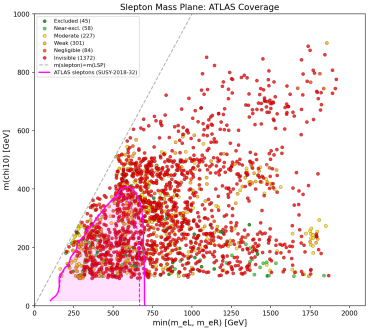
Caveat: SModelS v3.1.1 missing three key ATLAS analyses — true invisible fraction is lower.

Coverage details

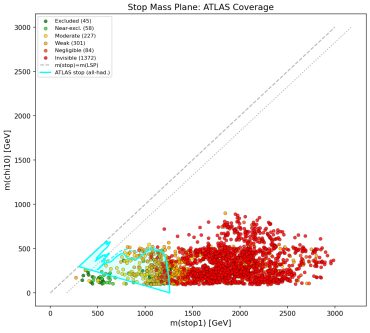
Mass Plane Scatter Plots



EWKino plane



Slepton plane



Stop plane

Many "invisible" models lie within published ATLAS contours — the gap is primarily in SModelS topology encoding, not ATLAS sensitivity.

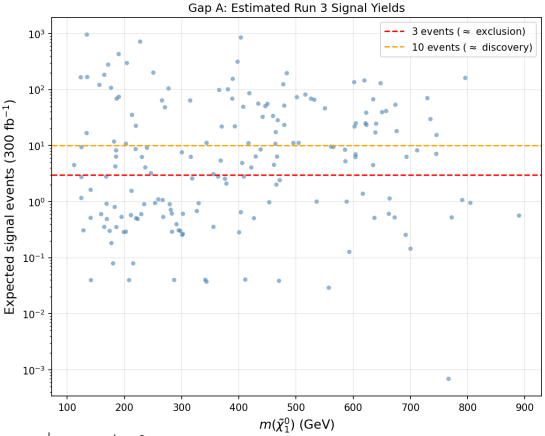
▸ Individual mass planes

Five ATLAS Blind Spots

Gap	Description	Models	Signal	Excl.	Disc.	Priority
A	Compressed Wino	520	Disapp. tracks + E_T^{miss}	102	72	**HIGH**
B	Compressed Higgsino	737	Soft dileptons + E_T^{miss}	42	19	**HIGH**
C	Light sleptons	509	Dilepton + E_T^{miss}	165	128	**MED**
D	Compressed stop	2	b-jet + displ. vtx	2	2	**MED**
E	Complex EWKino	80	Multi-lepton + E_T^{miss}	22	13	**LOW**

- 1,456 models (69.7%) at tier 0–1 (invisible/negligible)
- "Excl." $= N_{sig} > 3$, "Disc." $= N_{sig} > 10$ at $300\,\text{fb}^{-1}$
- **15 truly dark** models: invisible to both ATLAS and DARWIN
- 66 models with $c\tau(\tilde{\chi}_1^{\pm}) > 1\,\text{cm}$ (disappearing track candidates)

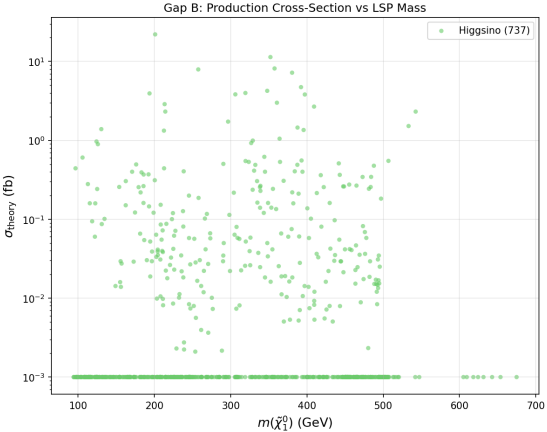
Gap A: Compressed Wino (520 models)

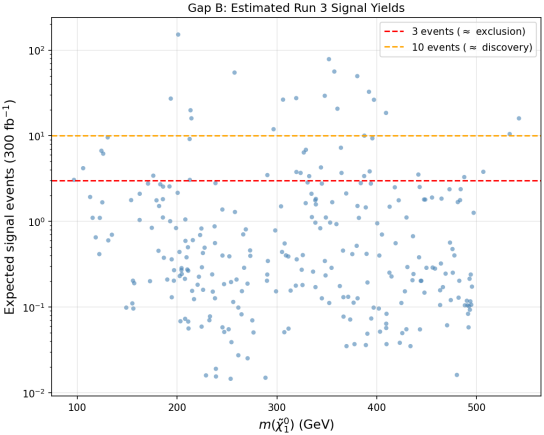


- All Wino LSP, median $\Delta m = 1.9\,\text{GeV}$. Signal: $\tilde{\chi}_1^\pm \tilde{\chi}_1^0 + \text{ISR}$, $\tilde{\chi}_1^\pm \to \pi^\pm \tilde{\chi}_1^0$ (displaced)
- 534 below ATLAS-SUSY-2018-19 limit ($\sim$660 GeV) but SModelS lacks topology
- Run 3 ITk $\to$ 800+ GeV; 102 excludable, 72 discoverable

▸ Gap A details

Gap B: Compressed Higgsino (737 models)

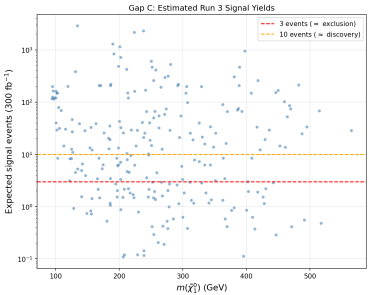




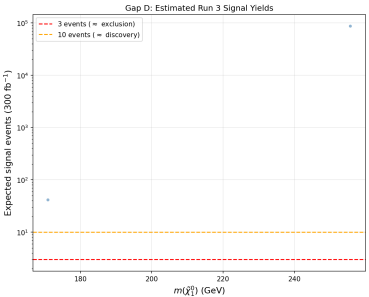
- All Higgsino LSP, median $\Delta m = 3.1$ GeV. Signal: $\tilde{\chi}_2^0\tilde{\chi}_1^\pm$ + ISR, soft dileptons
- $m_{\ell\ell}$ binning down to 1 GeV targets compressed spectrum
- Improved low-p_T e reco from ITk; 42 excludable, 19 discoverable

Gap B details

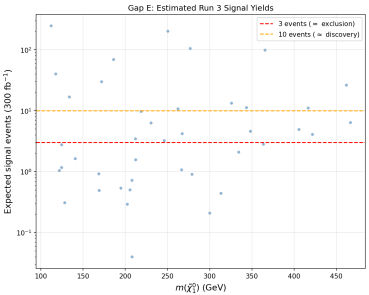
Gaps C / D / E: Sleptons, Stops, Complex EWKino



Gap C: Light sleptons (509)



Gap D: Compr. stop (2)



Gap E: Complex EWKino (80)

- **C:** 21 Bino+slepton models invisible despite ATLAS reach ∼700 GeV— pMSSM BR dilution
- **D:** Only 2 models but σ up to **14 pb**; long-lived $\tilde{\chi}_1^\pm$ falls between prompt & displaced
- **E:** Multi-body cascades; extend multi-lepton searches with asymmetric SRs

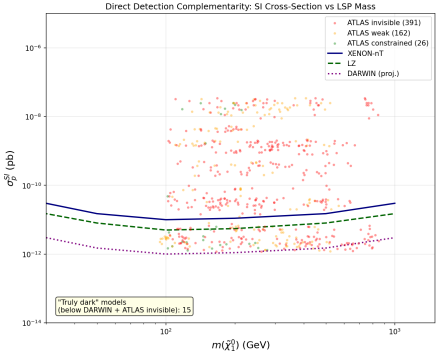
▸ Gap C/D/E details

Three Unexpected Blind Spots

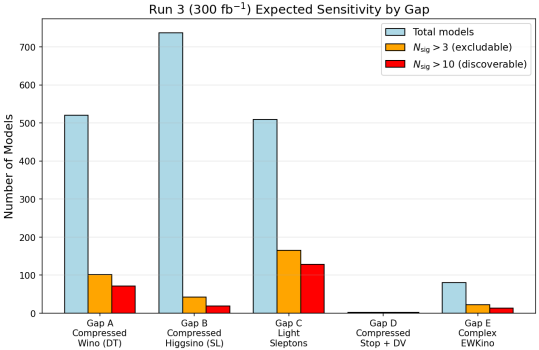
1. **Light sleptons with Bino LSP (21 models):** Sleptons at 270–580 GeV invisible despite ATLAS reach $\sim$700 GeV. pMSSM BRs dilute the clean dilepton signal; SModelS lacks this topology.
2. **Compressed stops with displaced charginos (2 models, up to 14 pb):** Stops within T2tt contour but Wino LSP creates long-lived $\tilde{\chi}_1^{\pm}$ ($c\tau \sim$ 1–10 cm). Falls between prompt and displaced searches — a **novel gap**.
3. **534 Wino models below disappearing-track limit:** ATLAS-SUSY-2018-19 excludes pure Winos to $\sim$660 GeV, but all invisible in SModelS. Encoding this **single analysis** would reclassify most "invisible" models.

Models at masses and σ where simplified-model results predict exclusion — the pMSSM reveals the gap.

Direct Detection & Run 3 Reach



SI σ_p vs m_{LSP}. **15 models** below DARWIN sensitivity AND invisible to ATLAS — can only be probed at LHC with dedicated searches.



Run 3 reach at 300 fb^{-1} across the five gaps. Combined: 333 excludable, 234 discoverable.

Benchmark models

Recommendations for ATLAS

1. **Encode existing searches in SModelS** (highest impact):
   Add SUSY-2018-19 (disapp. tracks), SUSY-2018-32 (sleptons), SUSY-2019-09 (compressed Higgsino).
   Would reclassify the majority of 1,372 "invisible" models.
2. **Extend disappearing-track search (Gap A):**
   Reach $\sim$460 $\rightarrow$ 800+ GeV with ITk. 102 models excludable.
3. **Lower soft-lepton thresholds (Gap B):**
   ISR-boosted with $m_{\ell\ell}$ binning to 1 GeV. 42 models excludable.
4. **Extend slepton reach (Gap C):**
   m_{T2}-based to 400–500 GeV. 165 models excludable.
5. **Novel displaced b-jet search (Gap D):**
   Displaced vertex + b-tagging. σ up to 14 pb.
6. **Multi-lepton cascade search (Gap E):**
   Asymmetric final-state SRs. 22 models excludable.

Summary

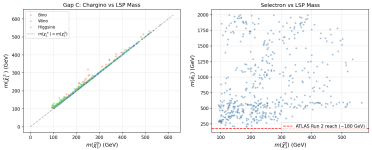
- **2,087 pMSSM models** survive all current constraints (from 26,048 evaluated)
- Viable pMSSM dominated by **compressed electroweakino spectra**: Wino 48%, Higgsino 48%, Bino 4%
- **69.7% invisible to ATLAS** in SModelS — but much of this is a SModelS encoding gap
- **Five blind spots** identified with concrete Run 3 search proposals and 30 benchmarks
- **15 truly dark models** invisible to both ATLAS and direct detection (DARWIN)
- Narrow corridors (Bino co-annihilation, compressed stop) intrinsically difficult — importance sampling is the best tool
- **Highest-impact action:** Encode 3 missing ATLAS analyses in SModelS

Full code, spectra, and analysis: `https://github.com/jwuerzinger/SUSY_agent`

Backup slides follow

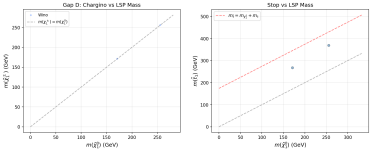
Backup: Kinematic Reach (Gaps C, D, E)

Gap C: Kinematic Reach



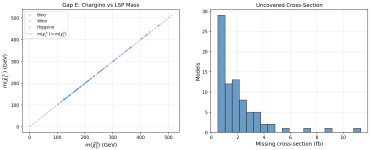
Gap C

Gap D: Kinematic Reach



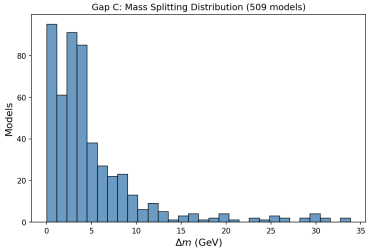
Gap D

Gap E: Kinematic Reach

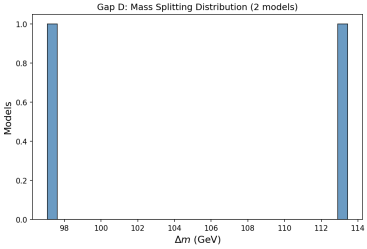


Gap E

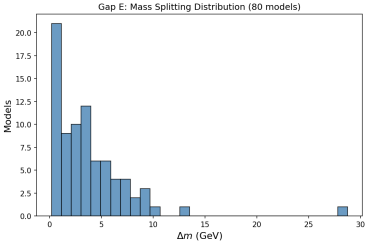
Backup: Mass Splitting Distributions (Gaps C, D, E)



Gap C

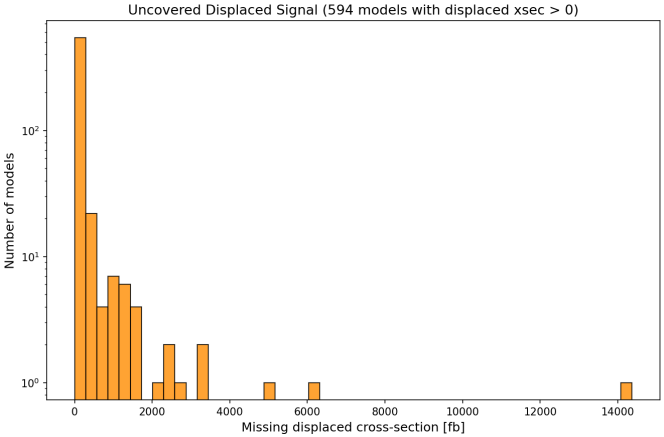


Gap D



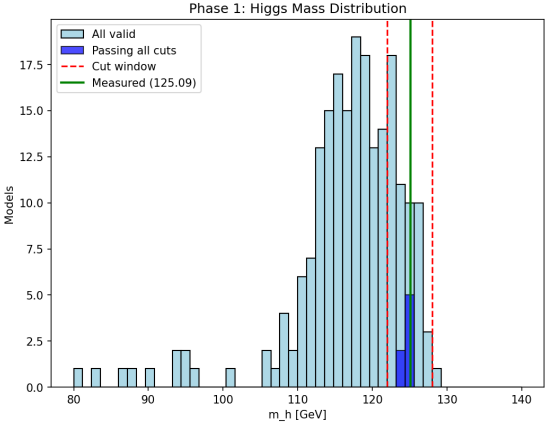
Gap E

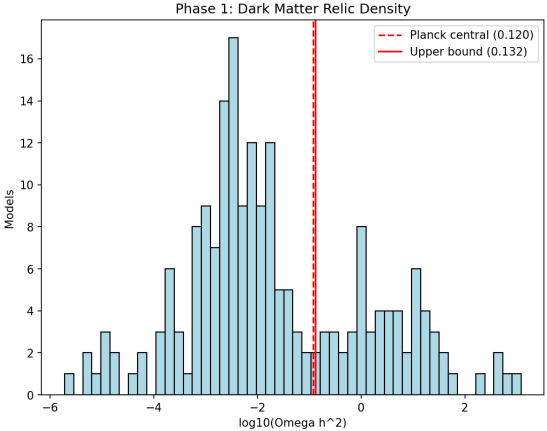
Backup: Displaced Cross-Section Distribution



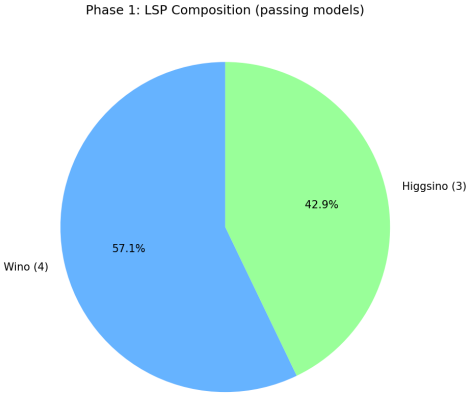
Production σ for models with displaced signatures ($c\tau > 1\,\mathrm{cm}$). 66 disappearing-track candidates.

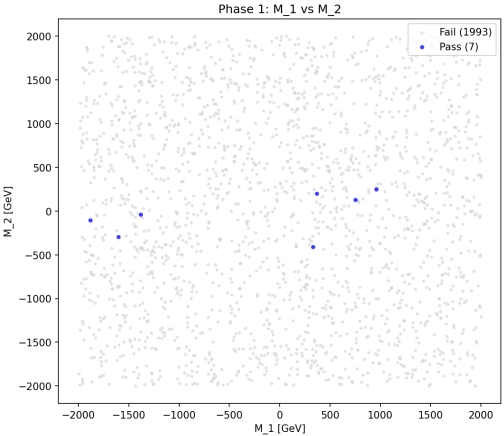
Backup: Phase 1 — Higgs Mass & Relic Density



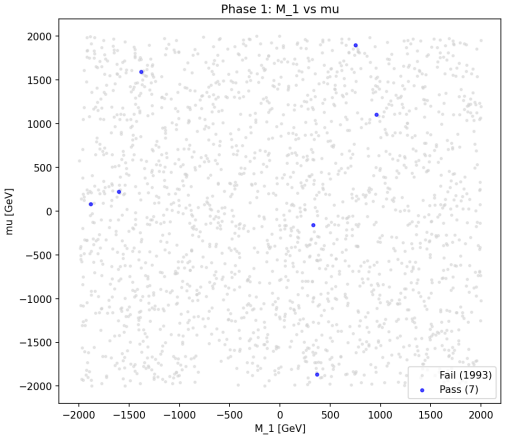


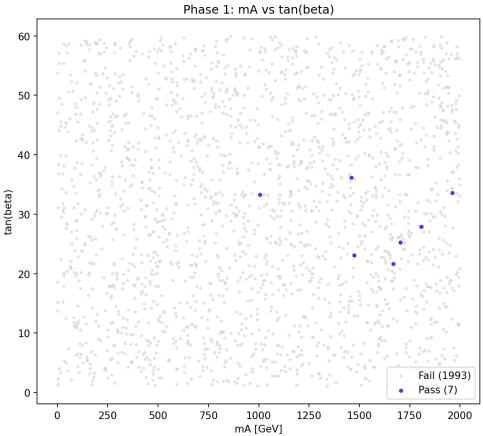
Backup: Phase 1 — LSP Composition & Parameter Space



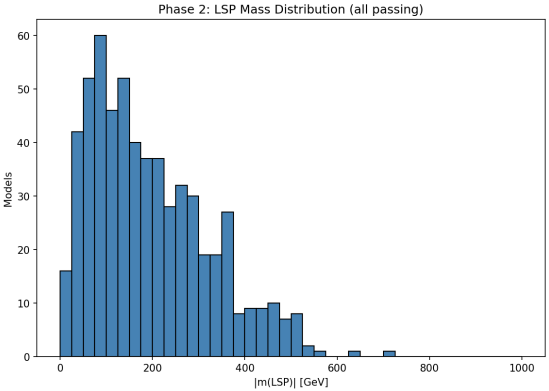


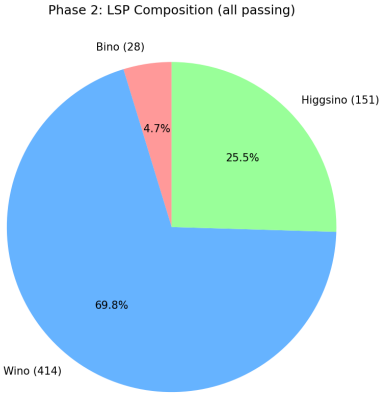
Backup: Phase 1 — M_1 vs μ and m_A vs $\tan\beta$



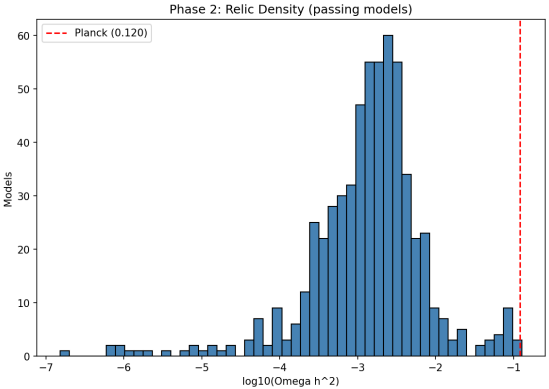


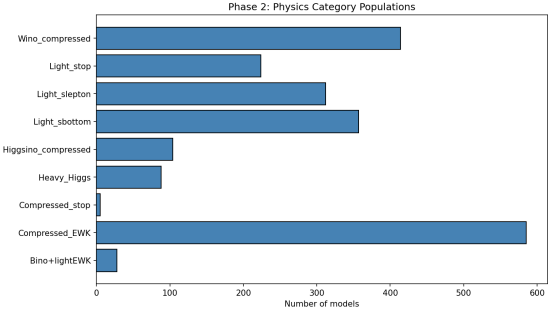
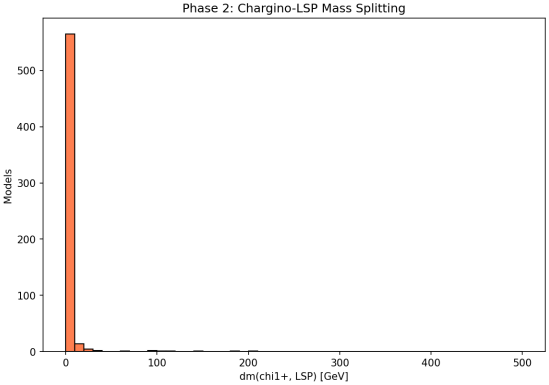
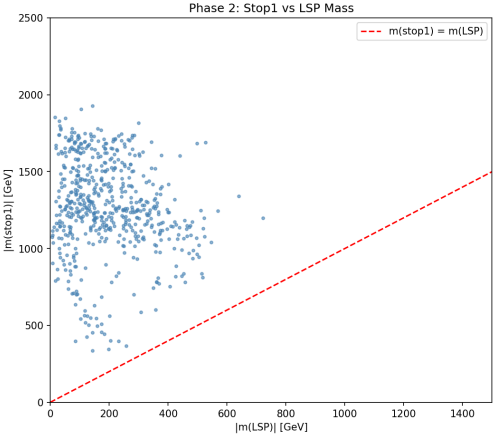
Backup: Phase 2 — LSP Mass & Composition





Backup: Phase 2 — Relic Density & Categories





Backup: Phase 2 — Chargino Δm & Stop vs LSP